

# IonShield — Data Sources

Every input is measured (NOAA / NASA / GFZ), forecaster-issued (NOAA), or explicitly operator-entered — and each output labels which. Nothing is synthetic. Base URLs are configurable (SWPC\_BASE\_URL, HAPI\_BASE\_URL) so an enclave can point at an internal mirror across a one-way diode instead of the public internet.

## Live operational feeds — NOAA SWPC, polled every 5 minutes

| Dataset                              | Endpoint (services.swpc.noaa.gov)              | Drives                                           |
|--------------------------------------|------------------------------------------------|--------------------------------------------------|
| Planetary Kp index (1-min)           | /json/planetary_k_index_1m.json                | Geomagnetic storm level — GPS, HF, scintillation |
| GOES X-ray flux 1–8 Å (6-hr)         | /json/goes/primary/xrays-6-hour.json           | Flare class — HF blackout (D-layer absorption)   |
| Solar wind plasma (2-hr)             | /products/solar-wind/plasma-2-hour.json        | Speed / density — storm driver context           |
| IMF magnetometer, Bz GSM (2-hr)      | /products/solar-wind/mag-2-hour.json           | Southward Bz — the storm trigger                 |
| GOES proton flux ≥10 MeV (3-day)     | /json/goes/primary/integral-protons-3-day.json | Polar cap absorption — poleward HF blackout      |
| NOAA 3-day Kp forecast               | /products/noaa-planetary-k-index-forecast.json | Forward risk + ATAK overlay time windows         |
| NOAA R/S/G scales + probabilities    | /products/noaa-scales.json                     | Forecaster-issued R/S/G probabilities            |
| F10.7 cm solar radio flux            | /json/f107_cm_flux.json                        | Ionospheric baseline (solar-activity proxy)      |
| GloTEC global Total Electron Content | /products/glotec/geojson_2d_urt.json           | Measured TEC — GPS error & radar range bias      |

Health checks report these as **noaa\_swpc** (7 feeds) + **ionosphere** (2 feeds). Each feed fails independently behind a circuit breaker; SHA-256 provenance is recorded on every value.

## Historical / backfill — on-demand, not live

| Dataset                                   | Source                                                                                    | Use                                                                                                        |
|-------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| NASA OMNI hourly merged (OMNI2_H0_MRG1HR) | NASA CDAWeb HAPI (cdaweb.gsfc.nasa.gov/hapi) — Bz GSM, density, velocity, proton flux, Kp | Backfills the local DB so the Kp ML forecaster has training history and storm scenarios can be precomputed |

## Replay datasets — recorded, for demos & validation

| Dataset                                                                                             | Source                                                                               | Use                                                                                     |
|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| May 2024 “Gannon” G5 storm — measured peak (Kp 9, Bz −50 nT, X5.8, 208 pfu) + GFZ 48-hr Kp timeline | NOAA SWPC G5 event reports; GOES-16 XRS + DSCOVR; GFZ Potsdam (kp.gfz.de, CC BY 4.0) | Labeled “REPLAY” demo scenarios — one real storm shown through different mission lenses |

## Operator-supplied — last-resort, disconnected ops

| Input                                                 | Source                                               | Use                                                                                              |
|-------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Manual Kp entry (+ optional flare class, proton flux) | Operator's S2 weather brief or space weather officer | When no live feed and no cache — runs the same doctrine rules, labeled MANUAL; expires after 3 h |

**Disconnected operation:** a pre-mission sync caches the full feed state (incl. the 3-day forecast); IonShield then serves that carried state in ADVISORY mode with no internet. Data quality degrades honestly with age — it never fabricates freshness.

[IonShield](#) · [PNT & communications mission assurance](#) · [live NOAA SWPC ingest](#) · [UDS/Zarf air-gap deployable](#)